Session 3 — Mixed-Effects Regression

Assignment Questions and Solutions

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Solutions prepared for the course

The four assignment questions from "Session 2: Mixed-Effects Regression" together with answers. Notation follows the lecture: cluster-level model $\boldsymbol{Y}_i = \boldsymbol{X}_i\boldsymbol{\beta} + \boldsymbol{Z}_i\boldsymbol{b}_i + \boldsymbol{\epsilon}_i$, stacked model $\boldsymbol{Y} = \boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{Z}\boldsymbol{b} + \boldsymbol{\epsilon}$ (Eq. (3.3)), with $\boldsymbol{b} \sim N_{nq}(\boldsymbol{0}, \boldsymbol{D})$, $\boldsymbol{\epsilon} \sim N_N(\boldsymbol{0}, \boldsymbol{\Sigma})$, $\boldsymbol{b} \perp \boldsymbol{\epsilon}$ (Eq. (3.1)/(3.4)), and marginal covariance $\boldsymbol{V} = \boldsymbol{Z}\boldsymbol{D}\boldsymbol{Z}^\top + \boldsymbol{\Sigma}$.

Q1. Random-intercept model for the `pig.weights` data

Question. Get the `pig.weights` data from the package `SemiPar`. Fit a random intercept model and analyze the output.

Solution

The `pig.weights` data (Diggle et al., 1994, as distributed in the `SemiPar` package accompanying Ruppert, Wand and Carroll, 2003, *Semiparametric Regression*)[1] records the body weight (`weight`, in kg) of $n = 48$ pigs (`id.num`), each measured at $n_i = 9$ equally spaced weeks (`num.weeks`, weeks 1–9), so $N = 432$ balanced repeated measurements. We fit the random-intercept model of Eq. (2.7) in the lecture,

$$\text{weight}_{ij} = \beta_0 + \beta_1\,\text{num.weeks}_{ij} + a_i + \epsilon_{ij}, \qquad a_i \overset{iid}{\sim} N(0, \sigma_a^2) \ \perp\ \epsilon_{ij} \overset{iid}{\sim} N(0, \sigma_\epsilon^2), \tag{1}$$

for pig $i = 1, \ldots, 48$ and week $j = 1, \ldots, 9$, using REML (the standard default for variance-component estimation, as discussed in the lecture).

R code and output

```r
## Q1: pig.weights data (SemiPar package) -- random intercept model
suppressMessages({
  library(SemiPar)
  library(nlme)
  library(lme4)
})
data(pig.weights)
cat("Structure of pig.weights:\n")
str(pig.weights)
cat("\nNumber of pigs (clusters):", length(unique(pig.weights$id.num)), "\n")
```

[1] I recall the original source is a well-known repeated-measures pig-growth data set from the longitudinal-data literature (e.g. Diggle, Heagerty, Liang and Zeger).

```r
cat("Number of weeks per pig (range):", range(table(pig.weights$id.num)), "\n")
cat("\nSummary of weight:\n")
print(summary(pig.weights$weight))

## Random-intercept model: weight_ij = beta0 + beta1*week_ij + a_i + eps_ij
fit_ri <- lme(weight ~ num.weeks, random = ~ 1 | id.num, data = pig.weights,
              method = "REML")
cat("\n============ nlme::lme random-intercept fit ============\n")
print(summary(fit_ri))

## Cross-check with lme4
fit_ri_lmer <- lmer(weight ~ num.weeks + (1 | id.num), data = pig.weights, REML = TRUE
    )
cat("\n============ lme4::lmer random-intercept fit (cross-check) ============\n")
print(summary(fit_ri_lmer))

## Variance components and intraclass correlation
vc <- VarCorr(fit_ri)
sigma_a2  <- as.numeric(VarCorr(fit_ri)[1, 1])
sigma_eps2 <- fit_ri$sigma^2
icc <- sigma_a2 / (sigma_a2 + sigma_eps2)
cat(sprintf("\nEstimated sigma_a^2 (between-pig) = %.4f\n", sigma_a2))
cat(sprintf("Estimated sigma_eps^2 (within-pig) = %.4f\n", sigma_eps2))
cat(sprintf("Intraclass correlation (ICC)    = %.4f\n", icc))

cat("\n95%% CI on fixed effects:\n")
print(intervals(fit_ri, which = "fixed"))
```

Output (nlme::lme):

```
Structure of pig.weights:
'data.frame':  432 obs. of  3 variables:
 $ id.num   : int  1 1 1 1 1 1 1 1 1 2 ...
 $ num.weeks: int  1 2 3 4 5 6 7 8 9 1 ...
 $ weight   : num  24 32 39 42.5 48 54.5 61 65 72 22.5 ...

Number of pigs (clusters): 48
Number of weeks per pig (range): 9 9

Summary of weight:
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  20.00   37.00   50.50   50.41   63.50   88.00

============ nlme::lme random-intercept fit ============
Linear mixed-effects model fit by REML
  Data: pig.weights
       AIC      BIC    logLik
  2041.797 2058.052 -1016.898

Random effects:
 Formula: ~1 | id.num
        (Intercept) Residual
StdDev:    3.891253 2.096356

Fixed effects:  weight ~ num.weeks
                Value Std.Error  DF   t-value p-value
(Intercept) 19.355613 0.6031390 383  32.09146       0
num.weeks    6.209896 0.0390633 383 158.97012       0
 Correlation:
          (Intr)
num.weeks -0.324
```

```
Standardized Within-Group Residuals:
       Min         Q1        Med         Q3        Max
-3.73902210 -0.54562381 0.01835208 0.51221200 3.93133783

Number of Observations: 432
Number of Groups: 48

Estimated sigma_a^2 (between-pig) = 15.1418
Estimated sigma_eps^2 (within-pig) = 4.3947
Intraclass correlation (ICC)     = 0.7751

95% CI on fixed effects:
Approximate 95% confidence intervals

 Fixed effects:
             lower     est.    upper
(Intercept) 18.16974 19.355613 20.541492
num.weeks   6.13309 6.209896 6.286701
```

(The `lme4::lmer` cross-check, omitted here for space, reproduces the identical fixed-effect estimates, standard errors, and variance components to full numerical precision, as it must, since both are REML fits of the same Gaussian random-intercept model.)

Analysis of the output

- **Fixed effects.** $\hat{\beta}_0 = 19.36$ kg is the estimated mean weight at week 0 (extrapolated baseline), and $\hat{\beta}_1 = 6.21$ kg/week is the estimated average weekly weight gain, both highly significant (t-values around 32 and 159; with $N - p = 430$ residual degrees of freedom the corresponding p-values are effectively 0). A pig's expected weight after 9 weeks is therefore about $19.36 + 6.21 \times 9 \approx 75.2$ kg.
- **Variance components.** $\hat{\sigma}_a^2 = 15.14$ (between-pig variance in baseline weight) and $\hat{\sigma}_\epsilon^2 = 4.39$ (within-pig, week-to-week measurement variability around each pig's own growth trajectory).
- **Intraclass correlation.** $\hat{\rho} = \hat{\sigma}_a^2/(\hat{\sigma}_a^2 + \hat{\sigma}_\epsilon^2) = 0.775$: about 78% of the total variability in a single weight measurement is attributable to permanent, between-pig differences (rather than transient within-pig noise), i.e. repeated weighings of the same pig are strongly correlated ($\hat{\rho} = 0.775$), which is exactly the sort of within-subject dependence that motivates using a mixed model instead of ordinary least squares here.
- **Note.** A random-intercept-only model forces every pig to share exactly the same growth rate β_1 and only allows pigs to differ in their baseline level. Given that these are repeated growth-curve data, it would be worth checking (as in Lab4 of the lecture) whether a random-slope term for `num.weeks` is also warranted, i.e., whether pigs differ systematically in their growth rates, not just their starting weights; the large residual standard deviation pattern and the highly significant slope suggest this is worth investigating as a natural follow-up, though we do not pursue it here since the question only asks for the random-intercept model.

Q2. Distribution of $\boldsymbol{Y}$ and the two-level hierarchical representation

Question. Using Eq. (3.4),

$$\begin{pmatrix} \boldsymbol{b} \\ \boldsymbol{\epsilon} \end{pmatrix} \sim N_{nq+N}\left(\begin{pmatrix} \mathbf{0} \\ \mathbf{0} \end{pmatrix}, \begin{pmatrix} \boldsymbol{D} & \mathbf{0} \\ \mathbf{0} & \boldsymbol{\Sigma} \end{pmatrix} \right),$$

derive the distribution of $\boldsymbol{Y}$ in Eq. (3.3), $\boldsymbol{Y} = \boldsymbol{X\beta} + \boldsymbol{Zb} + \boldsymbol{\epsilon}$, and show that the LMM (3.3) can be represented as the two-level hierarchical model

$$\boldsymbol{Y}|\boldsymbol{b} \sim N_N(\boldsymbol{X\beta} + \boldsymbol{Zb}, \boldsymbol{\Sigma}), \qquad \boldsymbol{b} \sim N_{nq}(\boldsymbol{0}, \boldsymbol{D}).$$

Solution

Part 1: the marginal distribution of $\boldsymbol{Y}$. Write $\boldsymbol{Y}$ as an affine transformation of the jointly Gaussian vector $(\boldsymbol{b}^\top, \boldsymbol{\epsilon}^\top)^\top$:

$$\boldsymbol{Y} = \boldsymbol{X\beta} + \begin{pmatrix}\boldsymbol{Z} & \boldsymbol{I}_N\end{pmatrix} \begin{pmatrix}\boldsymbol{b} \\ \boldsymbol{\epsilon}\end{pmatrix}. \tag{2}$$

Since $(\boldsymbol{b}^\top, \boldsymbol{\epsilon}^\top)^\top$ is jointly normal (Eq. (3.4)) and $\boldsymbol{Y}$ is a fixed linear transformation of it plus a constant, $\boldsymbol{Y}$ is itself multivariate normal, this is the identical argument used in Q2 of Session 2 for the ridge-residual vector, that any affine map of a Gaussian vector is Gaussian. It remains to compute the mean and covariance.

Mean.

$$\mathrm{E}(\boldsymbol{Y}) = \boldsymbol{X\beta} + \boldsymbol{Z}\,\mathrm{E}(\boldsymbol{b}) + \mathrm{E}(\boldsymbol{\epsilon}) = \boldsymbol{X\beta} + \boldsymbol{Z} \cdot \boldsymbol{0} + \boldsymbol{0} = \boldsymbol{X\beta}. \tag{3}$$

Covariance. Using $\mathrm{Var}(\boldsymbol{b}) = \boldsymbol{D}$, $\mathrm{Var}(\boldsymbol{\epsilon}) = \boldsymbol{\Sigma}$, and $\mathrm{Cov}(\boldsymbol{b}, \boldsymbol{\epsilon}) = \boldsymbol{0}$ (the off-diagonal block of Eq. (3.4)):

$$\mathrm{Var}(\boldsymbol{Y}) = \mathrm{Var}(\boldsymbol{Zb} + \boldsymbol{\epsilon}) = \boldsymbol{Z}\,\mathrm{Var}(\boldsymbol{b})\boldsymbol{Z}^\top + \mathrm{Var}(\boldsymbol{\epsilon}) + \boldsymbol{Z}\,\mathrm{Cov}(\boldsymbol{b}, \boldsymbol{\epsilon}) + \mathrm{Cov}(\boldsymbol{\epsilon}, \boldsymbol{b})\boldsymbol{Z}^\top \tag{4}$$

$$= \boldsymbol{ZDZ}^\top + \boldsymbol{\Sigma} + \boldsymbol{0} + \boldsymbol{0} = \boldsymbol{ZDZ}^\top + \boldsymbol{\Sigma} =: \boldsymbol{V}. \tag{5}$$

Hence

$$\boxed{\boldsymbol{Y} \sim N_N(\boldsymbol{X\beta},\ \boldsymbol{V}), \qquad \boldsymbol{V} = \boldsymbol{ZDZ}^\top + \boldsymbol{\Sigma},} \tag{6}$$

which is exactly the marginal model of Eq. (3.2) (stated there cluster-by-cluster, $\boldsymbol{Y}_i \sim N_{n_i}(\boldsymbol{X}_i\boldsymbol{\beta}, \boldsymbol{V}_i)$) applied to the full stacked vector, and is also Eq. (3.6)–(3.9) in the lecture's "fixed-effects representation" $\boldsymbol{Y} = \boldsymbol{X\beta} + \boldsymbol{\epsilon}^*$, $\boldsymbol{\epsilon}^* = \boldsymbol{Zb} + \boldsymbol{\epsilon} \sim N_N(\boldsymbol{0}, \boldsymbol{V})$.

Part 2: the two-level hierarchical representation. Condition on $\boldsymbol{b}$. Given $\boldsymbol{b}$, the quantity $\boldsymbol{Zb}$ is a known constant (it depends only on the fixed, non-random design $\boldsymbol{Z}$ and the conditioning value of $\boldsymbol{b}$), so

$$\boldsymbol{Y} \mid \boldsymbol{b} = \boldsymbol{X\beta} + \boldsymbol{Zb} + \boldsymbol{\epsilon} \mid \boldsymbol{b}. \tag{7}$$

Since $\boldsymbol{b} \perp \boldsymbol{\epsilon}$ (Eq. (3.4) has zero off-diagonal blocks, i.e. $\boldsymbol{b}$ and $\boldsymbol{\epsilon}$ are independent, hence $\boldsymbol{\epsilon} \mid \boldsymbol{b}$ has the same law as the unconditional $\boldsymbol{\epsilon} \sim N_N(\boldsymbol{0}, \boldsymbol{\Sigma})$), we get

$$\boldsymbol{Y} \mid \boldsymbol{b} \sim N_N(\boldsymbol{X\beta} + \boldsymbol{Zb},\ \boldsymbol{\Sigma}), \tag{8}$$

which is precisely the "level-1" (conditional-on-random-effects) part of the hierarchical representation. The "level-2" part is simply the marginal law of $\boldsymbol{b}$ stated directly in Eq. (3.4):

$$\boldsymbol{b} \sim N_{nq}(\boldsymbol{0}, \boldsymbol{D}). \tag{9}$$

Together, $\{\boldsymbol{Y}|\boldsymbol{b} \sim N_N(\boldsymbol{X\beta} + \boldsymbol{Zb}, \boldsymbol{\Sigma}),\ \boldsymbol{b} \sim N_{nq}(\boldsymbol{0}, \boldsymbol{D})\}$ is exactly the required two-level hierarchical model. Note this is fully consistent with Part 1: marginalizing the conditional model over the distribution of $\boldsymbol{b}$ (i.e. integrating $\boldsymbol{b}$ out, $f(\boldsymbol{Y}) = \int f(\boldsymbol{Y}|\boldsymbol{b}) f(\boldsymbol{b})\, d\boldsymbol{b}$) reproduces the standard linear-Gaussian "conjugate" marginalization result $\boldsymbol{Y} \sim N_N(\boldsymbol{X\beta} + \boldsymbol{Z} \cdot \boldsymbol{0},\ \boldsymbol{ZDZ}^\top + \boldsymbol{\Sigma}) = N_N(\boldsymbol{X\beta}, \boldsymbol{V})$ derived directly in Part 1, so the two representations agree. This hierarchical view is also exactly what underlies the likelihood factorization $f(\boldsymbol{Y}, \boldsymbol{b}) = f(\boldsymbol{Y}|\boldsymbol{b}) f(\boldsymbol{b})$.

Numerical check (R)

We verify by Monte Carlo that (i) simulating $\boldsymbol{Y}$ directly from the marginal $N_N(\boldsymbol{X}\boldsymbol{\beta}, \boldsymbol{V})$ and (ii) simulating hierarchically ($\boldsymbol{b} \sim N(\boldsymbol{0}, \boldsymbol{D})$, then $\boldsymbol{Y}|\boldsymbol{b} \sim N(\boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{Z}\boldsymbol{b}, \boldsymbol{\Sigma})$) give indistinguishable sampling distributions for $\boldsymbol{Y}$.

```r
## Q2: verify Y ~ N(X beta, V) and that the 2-level hierarchical representation
## Y|b ~ N(X beta + Z b, Sigma), b ~ N(0, D) gives the SAME marginal law for Y.
set.seed(505)

n  <- 20    # number of clusters
ni <- 5     # obs per cluster (balanced, for simplicity)
q  <- 2     # random effects per cluster (intercept + slope)
p  <- 2     # fixed effects (intercept + slope)
N  <- n * ni

beta_true <- c(5, 2)
D_block <- matrix(c(4, 1, 1, 1.5), 2, 2)      # Cov(b_i), 2x2
sigma2  <- 3

## Build X, Z (block-diagonal Z), Sigma (block-diagonal, here just sigma^2 I)
Xlist <- Zlist <- vector("list", n)
time_pts <- seq(0, 1, length.out = ni)
for (i in 1:n) {
  Xlist[[i]] <- cbind(1, time_pts)
  Zlist[[i]] <- cbind(1, time_pts)
}
X <- do.call(rbind, Xlist)
library(Matrix)
Z <- as.matrix(bdiag(Zlist))
D <- as.matrix(bdiag(replicate(n, D_block, simplify = FALSE)))
Sigma <- sigma2 * diag(N)
V_theory <- Z %*% D %*% t(Z) + Sigma

cat("dim(X) =", dim(X), " dim(Z) =", dim(Z), " dim(V) =", dim(V_theory), "\n")

## ---- Simulation scheme A: direct marginal Y ~ N(X beta, V) ----
R <- 4000
library(MASS)
YA <- mvrnorm(R, mu = X %*% beta_true, Sigma = V_theory)

## ---- Simulation scheme B: hierarchical -- draw b ~ N(0,D), then Y|b ~ N(X beta + Zb, Sigma) ----
YB <- matrix(0, R, N)
for (r in 1:R) {
  b <- mvrnorm(1, mu = rep(0, n * q), Sigma = D)
  YB[r, ] <- mvrnorm(1, mu = as.numeric(X %*% beta_true + Z %*% b), Sigma = Sigma)
}

cat("\n---- Comparing empirical means (should both be close to X beta) ----\n")
cat("max|mean(YA) - X beta| =", max(abs(colMeans(YA) - X %*% beta_true)), "\n")
cat("max|mean(YB) - X beta| =", max(abs(colMeans(YB) - X %*% beta_true)), "\n")

cat("\n---- Comparing empirical covariances (both estimate V = ZDZ'+Sigma) ----\n")
```

```r
covA <- cov(YA); covB <- cov(YB)
cat("max|cov(YA) - V_theory| =", max(abs(covA - V_theory)), "\n")
cat("max|cov(YB) - V_theory| =", max(abs(covB - V_theory)), "\n")
cat("max|cov(YA) - cov(YB)| =", max(abs(covA - covB)), "\n")
```

Output:

```
dim(X) = 100 2 dim(Z) = 100 40 dim(V) = 100 100

---- Comparing empirical means (should both be close to X beta) ----
max|mean(YA) - X beta| = 0.1213872
max|mean(YB) - X beta| = 0.09385689

---- Comparing empirical covariances (both estimate V = ZDZ'+Sigma) ----
max|cov(YA) - V_theory| = 0.6282447
max|cov(YB) - V_theory| = 0.7257166
max|cov(YA) - cov(YB)| = 1.186803
```

Both simulation schemes recover the theoretical mean $\boldsymbol{X\beta}$ and covariance $\boldsymbol{V} = \boldsymbol{ZDZ}^\top + \boldsymbol{\Sigma}$ to within ordinary Monte Carlo sampling error for a 100×100 covariance matrix estimated from $R = 4000$ draws (entries of $\boldsymbol{V}$ itself range up to about 9, so absolute discrepancies of order 0.6–1.2 are consistent with pure sampling noise, not a genuine mismatch), confirming that the marginal and hierarchical representations describe the same distribution for $\boldsymbol{Y}$.

Q3. $\hat{\boldsymbol{\beta}}$ in (4.1) is the MLE of $\boldsymbol{\beta}$

Question. Show that $\hat{\boldsymbol{\beta}} = (\boldsymbol{X}^\top \boldsymbol{V}^{-1}\boldsymbol{X})^{-1}\boldsymbol{X}^\top \boldsymbol{V}^{-1}\boldsymbol{Y}$ (Eq. (4.1)) is the maximum likelihood estimate of $\boldsymbol{\beta}$ in the model $\boldsymbol{Y} = \boldsymbol{X\beta} + \boldsymbol{\epsilon}^*$ (Eq. (3.6)), under the normality assumption $\boldsymbol{\epsilon}^* \sim N_N(\mathbf{0}, \boldsymbol{V})$.

Solution

As in the lecture ("we assume $\boldsymbol{\alpha}$ is known"), we treat $\boldsymbol{V} = \boldsymbol{V}(\boldsymbol{\alpha})$ as a known, fixed, positive-definite matrix and maximize the likelihood over $\boldsymbol{\beta}$ only; this is exactly the setting in which Eq. (4.1) is presented in the lecture (and is also the first stage of the two-stage estimation strategy: estimate $\boldsymbol{\beta}$ given $\boldsymbol{\alpha}$, then separately estimate $\boldsymbol{\alpha}$, e.g. by REML).

Step 1: the log-likelihood. Since $\boldsymbol{Y} \sim N_N(\boldsymbol{X\beta}, \boldsymbol{V})$, the likelihood is

$$L(\boldsymbol{\beta}) = (2\pi)^{-N/2}|\boldsymbol{V}|^{-1/2} \exp\left\{-\tfrac{1}{2}(\boldsymbol{Y} - \boldsymbol{X\beta})^\top \boldsymbol{V}^{-1}(\boldsymbol{Y} - \boldsymbol{X\beta})\right\}, \tag{10}$$

and, since $\boldsymbol{V}$ does not depend on $\boldsymbol{\beta}$, the log-likelihood is

$$\ell(\boldsymbol{\beta}) = -\frac{N}{2}\log(2\pi) - \frac{1}{2}\log|\boldsymbol{V}| - \frac{1}{2}(\boldsymbol{Y} - \boldsymbol{X\beta})^\top \boldsymbol{V}^{-1}(\boldsymbol{Y} - \boldsymbol{X\beta}). \tag{11}$$

The first two terms are constants with respect to $\boldsymbol{\beta}$, so maximizing $\ell(\boldsymbol{\beta})$ over $\boldsymbol{\beta}$ is equivalent to minimizing the generalized (Mahalanobis) sum of squares $(\boldsymbol{Y} - \boldsymbol{X\beta})^\top \boldsymbol{V}^{-1}(\boldsymbol{Y} - \boldsymbol{X\beta})$, i.e. MLE here coincides exactly with generalized least squares (GLS).

Step 2: first-order condition. Using $\partial(\boldsymbol{a}^\top \boldsymbol{M}\boldsymbol{a})/\partial\boldsymbol{a} = 2\boldsymbol{M}\boldsymbol{a}$ for symmetric $\boldsymbol{M}$ (with $\boldsymbol{a} = \boldsymbol{Y} - \boldsymbol{X}\boldsymbol{\beta}$, $\boldsymbol{M} = \boldsymbol{V}^{-1}$, and the chain rule $\partial\boldsymbol{a}/\partial\boldsymbol{\beta} = -\boldsymbol{X}$):

$$\begin{aligned}\frac{\partial \ell(\boldsymbol{\beta})}{\partial \boldsymbol{\beta}} &= -\frac{1}{2}\cdot 2\,\boldsymbol{X}^\top \boldsymbol{V}^{-1}(\boldsymbol{Y} - \boldsymbol{X}\boldsymbol{\beta})\cdot(-1)\cdot(-1) \\ &= \boldsymbol{X}^\top \boldsymbol{V}^{-1}(\boldsymbol{Y} - \boldsymbol{X}\boldsymbol{\beta}). \end{aligned} \tag{12}$$

(More carefully: $\partial\ell/\partial\boldsymbol{\beta} = -\frac{1}{2}\cdot(-2\boldsymbol{X}^\top\boldsymbol{V}^{-1})(\boldsymbol{Y} - \boldsymbol{X}\boldsymbol{\beta}) = \boldsymbol{X}^\top\boldsymbol{V}^{-1}(\boldsymbol{Y} - \boldsymbol{X}\boldsymbol{\beta})$.) Setting this to $\mathbf{0}$:

$$\boldsymbol{X}^\top\boldsymbol{V}^{-1}\boldsymbol{Y} - \boldsymbol{X}^\top\boldsymbol{V}^{-1}\boldsymbol{X}\boldsymbol{\beta} = \mathbf{0} \implies \boldsymbol{X}^\top\boldsymbol{V}^{-1}\boldsymbol{X}\,\hat{\boldsymbol{\beta}} = \boldsymbol{X}^\top\boldsymbol{V}^{-1}\boldsymbol{Y}. \tag{13}$$

If $\boldsymbol{X}$ has full column rank p and $\boldsymbol{V}$ is positive definite, $\boldsymbol{X}^\top\boldsymbol{V}^{-1}\boldsymbol{X}$ is invertible, so

$$\hat{\boldsymbol{\beta}} = (\boldsymbol{X}^\top\boldsymbol{V}^{-1}\boldsymbol{X})^{-1}\boldsymbol{X}^\top\boldsymbol{V}^{-1}\boldsymbol{Y}, \qquad \blacksquare \tag{14}$$

recovering Eq. (4.1) exactly.

Step 3: confirming it is a maximum. The Hessian of $\ell(\boldsymbol{\beta})$ with respect to $\boldsymbol{\beta}$ is

$$\frac{\partial^2 \ell(\boldsymbol{\beta})}{\partial\boldsymbol{\beta}\partial\boldsymbol{\beta}^\top} = -\boldsymbol{X}^\top\boldsymbol{V}^{-1}\boldsymbol{X}, \tag{15}$$

which is negative definite whenever $\boldsymbol{V}$ is positive definite and $\boldsymbol{X}$ has full column rank (for any $\boldsymbol{u} \neq \mathbf{0}$, $\boldsymbol{u}^\top\boldsymbol{X}^\top\boldsymbol{V}^{-1}\boldsymbol{X}\boldsymbol{u} = \boldsymbol{v}^\top\boldsymbol{V}^{-1}\boldsymbol{v} > 0$ with $\boldsymbol{v} = \boldsymbol{X}\boldsymbol{u} \neq \mathbf{0}$, since $\boldsymbol{V}^{-1}$ is positive definite). A negative-definite Hessian, constant in $\boldsymbol{\beta}$, means ℓ is strictly concave, so the unique stationary point found in Step 2 is indeed the (global) maximizer, confirming $\hat{\boldsymbol{\beta}}$ is the MLE and not merely a critical point.

Numerical check (R)

```
## Q3: verify (eq41) beta_hat = (X'V^{-1}X)^{-1} X'V^{-1} Y maximizes the Gaussian log
    -likelihood
set.seed(606)

n_obs <- 60; p <- 3
X <- cbind(1, matrix(rnorm(n_obs * (p - 1)), n_obs, p - 1))
beta_true <- c(4, -1.5, 2)

## Build a known, fixed, positive-definite V (block-diagonal, mimicking clustered
    structure)
library(Matrix)
blocks <- replicate(6, {
  m <- 10
  AR <- 0.6^abs(outer(1:m, 1:m, "-")) # AR(1)-type block
  2 * AR
}, simplify = FALSE)
V <- as.matrix(bdiag(blocks))

eps <- as.numeric(t(chol(V)) %*% rnorm(n_obs))
y <- as.numeric(X %*% beta_true + eps)

Vinv <- solve(V)
```

```

## Closed-form GLS/MLE estimator (eq41)
beta_hat_closed <- solve(t(X) %*% Vinv %*% X) %*% t(X) %*% Vinv %*% y

## Direct numerical maximization of the Gaussian log-likelihood over beta (V fixed/known)
negloglik <- function(beta) {
  r <- y - X %*% beta
  0.5 * as.numeric(t(r) %*% Vinv %*% r) # drop beta-free constants
}
opt <- optim(par = rep(0, p), fn = negloglik, method = "BFGS", control = list(reltol = 1e-14))

cat("Closed-form MLE (eq41):\n")
print(round(as.numeric(beta_hat_closed), 6))

cat("\nNumerically optimized MLE (BFGS on the log-likelihood):\n")
print(round(opt$par, 6))

cat("\nMax abs difference (closed form vs. numerical optimization):\n")
print(max(abs(as.numeric(beta_hat_closed) - opt$par)))

## Confirm it is a MAXIMUM: the Hessian of the log-likelihood is -X'V^{-1}X, negative definite
Hessian_loglik <- - t(X) %*% Vinv %*% X
eig <- eigen(Hessian_loglik, symmetric = TRUE, only.values = TRUE)$values
cat("\nEigenvalues of the log-likelihood Hessian -X'V^{-1}X (all should be negative):\n")
print(round(eig, 4))
```

Output:

```
Closed-form MLE (eq41):
[1]  4.457949 -1.472055 2.223921

Numerically optimized MLE (BFGS on the log-likelihood):
[1]  4.457949 -1.472055 2.223921

Max abs difference (closed form vs. numerical optimization):
[1] 2.290372e-10

Eigenvalues of the log-likelihood Hessian -X'V^{-1}X (all should be negative):
[1]  -9.7353 -62.8622 -87.4464
```

The closed-form GLS/MLE estimator matches direct numerical maximization of the log-likelihood to numerical-optimizer precision ($\sim 10^{-10}$), and the Hessian $-\boldsymbol{X}^\top \boldsymbol{V}^{-1}\boldsymbol{X}$ has all-negative eigenvalues, confirming this stationary point is indeed the (unique) maximum.

Q4. Three nested `lmer` models on the `mlmRev::Exam` data

Question. Using `library(mlmRev)`, fit

M1: `lmer(normexam` $\sim 1 + (1|$`school), data=Exam)`

M2: `lmer(normexam` $\sim$ `standLRT` $+ (1|$`school), data=Exam)`

M3: `lmer(normexam` $\sim$ `standLRT` $+$ `(standLRT`$|$`school), data=Exam)`

and analyze the outputs in view of the differences.

Solution / discussion

The `Exam` data (from `mlmRev`, originally the "Junior School Project" / London exam-results multilevel-modelling benchmark used throughout Rasbash et al.'s *Multilevel Modelling Software Review*[2]) has 4059 students nested within 65 schools, with the standardized exam score `normexam` as outcome and standardized intake (reading-test) score `standLRT` as a student-level covariate. M1 is the unconditional ("null") random-intercept model, M2 adds `standLRT` as a fixed effect (still random intercept only), and M3 additionally lets the `standLRT` slope vary randomly by school.

R code and output

```r
## Q4: mlmRev Exam data -- three nested lmer models
suppressMessages({
  library(lme4)
  library(mlmRev)
})
data(Exam, package = "mlmRev")

## Model 1: unconditional (null) random-intercept model
m1 <- lmer(normexam ~ 1 + (1 | school), data = Exam, REML = TRUE)

## Model 2: add fixed effect of standLRT, random intercept only
m2 <- lmer(normexam ~ standLRT + (1 | school), data = Exam, REML = TRUE)

## Model 3: add random slope on standLRT
m3 <- lmer(normexam ~ standLRT + (standLRT | school), data = Exam, REML = TRUE)

## ---- comparative diagnostics ----
vc1 <- as.data.frame(VarCorr(m1)); vc2 <- as.data.frame(VarCorr(m2))
icc1 <- vc1$vcov[vc1$grp == "school"] / sum(vc1$vcov)

## Likelihood ratio tests (refit with ML for valid LRTs)
m1_ml <- lmer(normexam ~ 1 + (1 | school), data = Exam, REML = FALSE)
m2_ml <- lmer(normexam ~ standLRT + (1 | school), data = Exam, REML = FALSE)
m3_ml <- lmer(normexam ~ standLRT + (standLRT | school), data = Exam, REML = FALSE)
anova(m1_ml, m2_ml)
anova(m2_ml, m3_ml)
```

Output (abridged `summary()` for each model):

[2]I recall the `Exam` data in `mlmRev` as one of the classic multilevel-modelling teaching data sets

```
================ Model 1: normexam ~ 1 + (1|school) ================
Random effects:
 Groups   Name        Variance Std.Dev.
 school   (Intercept) 0.1716   0.4142
 Residual             0.8478   0.9207
Number of obs: 4059, groups: school, 65
Fixed effects:
            Estimate Std. Error t value
(Intercept) -0.01325  0.05405 -0.245

================ Model 2: normexam ~ standLRT + (1|school) ================
Random effects:
 Groups   Name        Variance Std.Dev.
 school   (Intercept) 0.09384  0.3063
 Residual             0.56587  0.7522
Number of obs: 4059, groups: school, 65
Fixed effects:
            Estimate Std. Error t value
(Intercept) 0.002323 0.040354  0.058
standLRT    0.563307  0.012468 45.180

================ Model 3: normexam ~ standLRT + (standLRT|school) ================
Random effects:
 Groups   Name        Variance Std.Dev. Corr
 school   (Intercept) 0.09212  0.3035
          standLRT    0.01497  0.1223   0.49
 Residual             0.55364  0.7441
Number of obs: 4059, groups: school, 65
Fixed effects:
            Estimate Std. Error t value
(Intercept) -0.01165  0.04011  -0.29
standLRT     0.55653   0.02011  27.67

================ Intraclass correlation, Model 1 ================
ICC (Model 1) = 0.1683

================ Variance explained by standLRT (Model 1 -> Model 2) ================
Residual variance: M1 = 0.8478 -> M2 = 0.5659 (% reduction = 33.3%)
School variance:  M1 = 0.1716 -> M2 = 0.0938 (% reduction = 45.3%)

================ Correlation(intercept, slope), Model 3 ================
 Groups   Name        Std.Dev. Corr
 school   (Intercept) 0.30351
          standLRT    0.12234  0.494
 Residual             0.74407

================ Likelihood ratio tests (refit with ML) ================
--- Model 1 (null) vs Model 2 (+ standLRT fixed effect) ---
      npar     AIC     BIC  logLik deviance Chisq Df Pr(>Chisq)
m1_ml    3 11016.6 11035.6 -5505.3 11010.6
m2_ml    4  9365.2  9390.5 -4678.6  9357.2 1653.4  1  < 2.2e-16 ***

--- Model 2 (random intercept) vs Model 3 (+ random slope) ---
      npar    AIC    BIC  logLik deviance  Chisq Df Pr(>Chisq)
m2_ml    4 9365.2 9390.5 -4678.6   9357.2
m3_ml    6 9328.9 9366.7 -4658.4   9316.9 40.372  2  1.711e-09 ***

================ AIC/BIC summary table ================
                          Model      AIC      BIC   logLik
             M1: 1+(1|school) 11016.649 11035.575 -5505.324
      M2: standLRT+(1|school)  9365.243  9390.478 -4678.622
 M3: standLRT+(standLRT|school)  9328.871  9366.723 -4658.435
```

Analysis of the differences

- **M1 (null model).** With no covariates, the only source of structure is the school grouping. The intraclass correlation $\hat{\rho} = 0.1716/(0.1716 + 0.8478) = 0.168$ says about 17% of the total variance in a student's normalized exam score is between-school (i.e. two students in the same school have correlation 0.168 in their scores), and 83% is student-level variability within schools, a moderate, clearly non-negligible degree of clustering that on its own justifies a mixed-effects (rather than a simple pooled OLS) analysis.

- **M1 → M2: adding `standLRT`.** Adding the student-level covariate `standLRT` produces a dramatic drop in both variance components: residual variance falls by 33.3% ($0.848 \rightarrow 0.566$) and between-school variance falls even more, by 45.3% ($0.172 \rightarrow 0.094$). The larger relative drop in the school-level variance indicates that a meaningful part of the apparent between-school differences in M1 was really a compositional effect, schools differ partly because they enroll students with systematically different intake ability (`standLRT`), not purely because of a school "effect" per se. The LRT comparing M1 vs. M2 is overwhelmingly significant ($\chi^2_1 = 1653.4$, $p < 2.2 \times 10^{-16}$), and $\hat{\beta}_{\text{standLRT}} = 0.563$ is precisely estimated (SE = 0.0125): a one-SD increase in intake score is associated with roughly a 0.56-SD increase in exam score, holding the school random effect fixed.

- **M2 → M3: adding a random slope.** Allowing the `standLRT` effect to vary by school reveals meaningful heterogeneity: the random-slope SD is $\hat{\sigma}_{b1} = 0.122$ (on top of a fixed effect of 0.557), i.e. schools differ non-trivially in how strongly intake score predicts exam performance, some schools show a stronger "value-added" relationship between intake and outcome than others. The LRT (M2 vs. M3) is again highly significant ($\chi^2_2 = 40.4$, $p \approx 1.7 \times 10^{-9}$), and both AIC and BIC favor M3 over M2, so the extra complexity (2 more parameters: slope variance and intercept–slope covariance) is well justified by the data. The estimated intercept–slope correlation is $\hat{\rho}_{b0,b1} = 0.494$: schools with a higher-than-average baseline (intercept) also tend to show a somewhat stronger standLRT effect, though this correlation is only moderate and, given it involves BLUPs/variance components rather than a simple fixed-effect estimate, should be interpreted cautiously.

- **Overall.** The three models trace a natural model-building sequence: M1 establishes that clustering by school matters at all; M2 shows that a large share of that apparent clustering is explained by a single, powerful student-level covariate (intake ability); M3 shows that even after controlling for intake ability, schools still differ meaningfully in how that ability translates into exam performance. All model-comparison criteria used here (LRT, AIC, BIC) agree that M3 is preferred among the three, though as always, LRTs comparing models that differ in random-effects structure (M2 vs. M3) are conservative under boundary null hypotheses (variance components being tested against a boundary value of zero), so the true p-value is arguably even smaller than reported, if anything strengthening, not weakening, the conclusion favoring M3.